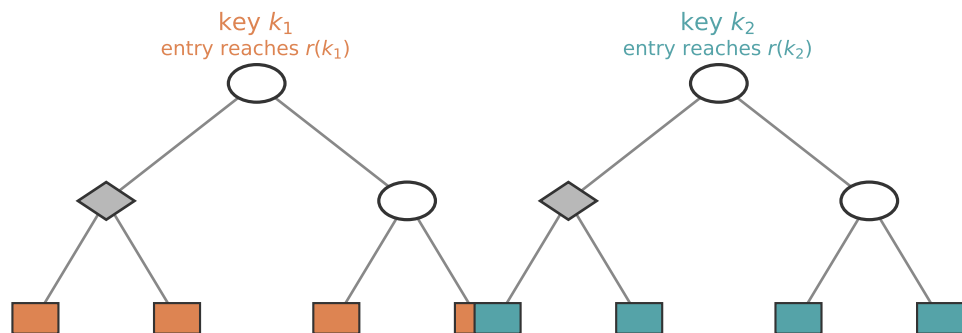


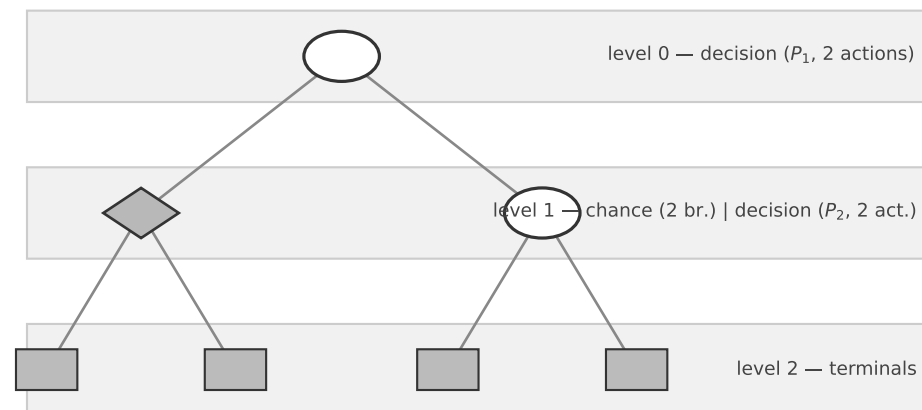
(a) The population: same topology, different data



decision node (circle) chance node (diamond) terminal payoffs (colored squares)

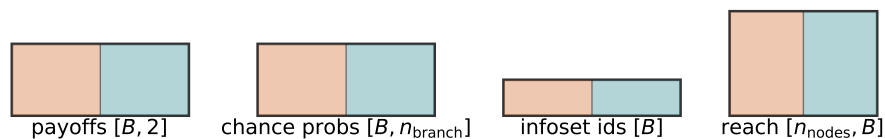
identical shape; only the colored per-instance data differ

(b) Topology bucketing: group nodes by level and shape

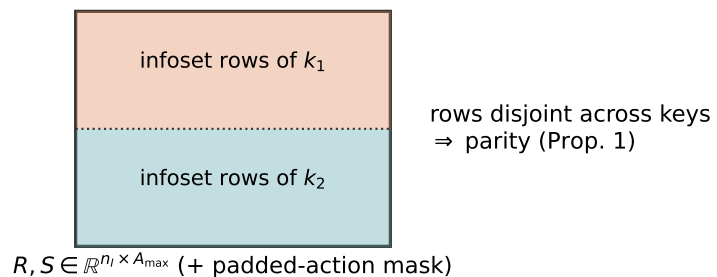


forward groups: children binned by parent kind;
backward groups: (player, n_{actions}) or (chance, n_{branch})

(c) Compiled structure-of-arrays: population = trailing axis



columns = instances (B); the two subgames' data are interleaved columns

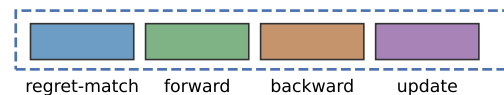


(d) One CFR+ iteration: two arms, same kernels

sequential (per-tree): host loop over keys



fused: one launch sequence; K folded into the tensor dimension



same four kernels in both arms; the only difference is batching granularity. At G4: 12 launch sequences per iteration $\rightarrow$ 1; fused time stays flat (197.6 $\rightarrow$ 202.4 ms, N : 1 $\rightarrow$ 12).